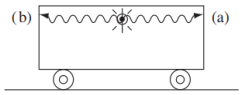
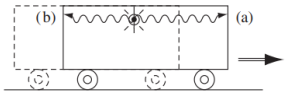
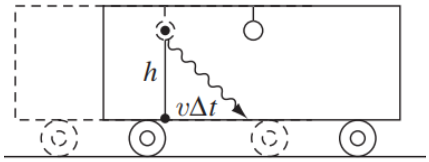
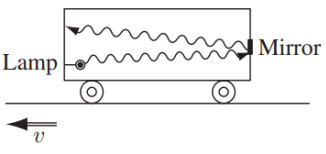
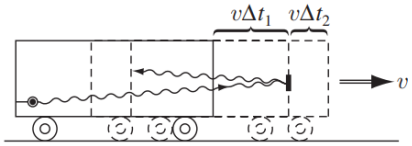
H12: Electrodynamics and relativity

12.1 the special theory of relativity

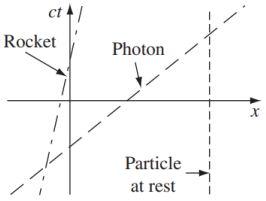
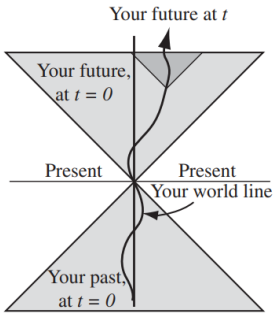
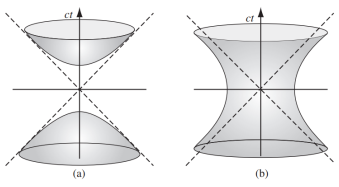
12.1.1 Einstein's postulates

principle of relativity	= the laws of physics apply in any inertial reference frame ie: in all frames at rest or at cte velocity
Einstein's postulates	1: principle of relativity 2: universal speed of light: speed of light in a vacuum is the same for all inertial observers
Einstein's velocity addition rule	Suppose A is on a train B going at a speed v_{BC} with respect to the ground C > if i walk at a pace v_{AB} with respect to the train, my speed with respect to the ground is: $$v_{AC} = \frac{v_{AB} + v_{BC}}{1 + (v_{AB}v_{BC}/c^2)}$$

12.1.2 the geometry of relativity

the relativity of simultaneity	Two events that are simultaneous in one inertial system er not simultaneous in another ex: light source in a fast moving train:  FIGURE 12.4  FIGURE 12.5
time dilation	time goes slower in moving objects ex: consider a light ray in the train that strikes the floor of the train > for the observer on the train the time it takes for the light to hit the floor is: $$\Delta\bar{t} = \frac{h}{c}.$$ for the observer on the ground, the same ray must travel farther: $$\Delta t = \frac{\sqrt{h^2 + (v\Delta t)^2}}{c}$$ thus we have: $$\Delta\bar{t} = \sqrt{1 - v^2/c^2}\,\Delta t.$$ also define the quantity: $\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}$. 
Lorentz contraction	Moving object are shortened only in the direction of travel ex: light ray travels through a train and bounces of a mirror at the end > to an observer on the train, the time it takes is: $$\Delta\bar{t} = 2\frac{\Delta\bar{x}}{c}$$ to an observer on the ground, Δt_1 is the time it takes to get to the front and Δt_2 back again: $$\Delta t_1 = \frac{\Delta x + v\Delta t_1}{c}, \quad \Delta t_2 = \frac{\Delta x - v\Delta t_2}{c},$$ or thus: $$\Delta t_1 = \frac{\Delta x}{c - v}, \quad \Delta t_2 = \frac{\Delta x}{c + v}.$$ and the total time is thus: $\Delta\bar{t} = \sqrt{1 - v^2/c^2}\,\Delta t.$ we conclude: $$\Delta\bar{x} = \frac{1}{\sqrt{1 - v^2/c^2}}\Delta x.$$  

12.1.3 the Lorentz transformations	
Galilean transformations	In classical mechanics, events are simultaneous for different observers > they will measure the same time and dimensions nl: we don't account for the speed of light > then we have the transformations: (i) $\bar{x} = x - vt,$ (ii) $\bar{y} = y,$ (iii) $\bar{z} = z,$ (iv) $\bar{t} = t.$
Lorentz transformations	When we take c as the maximum velocity, we get the transformations: (i) $\bar{x} = \gamma(x - vt),$ (ii) $\bar{y} = y,$ (iii) $\bar{z} = z,$ (iv) $\bar{t} = \gamma\left(t - \frac{v}{c^2}x\right).$ with $\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}}.$
12.1.4 the structure of space-time	
Four-vector notation	Define: $x^0 \equiv ct, \quad \beta \equiv \frac{v}{c}.$ and now set: $x^1 = x, \quad x^2 = y, \quad x^3 = z,$ then the Lorentz-transformations read: $\bar{x}^0 = \gamma(x^0 - \beta x^1),$ $\bar{x}^1 = \gamma(x^1 - \beta x^0),$ $\bar{x}^2 = x^2,$ $\bar{x}^3 = x^3.$ or thus in matrix form: $\begin{pmatrix} \bar{x}^0 \\ \bar{x}^1 \\ \bar{x}^2 \\ \bar{x}^3 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x^0 \\ x^1 \\ x^2 \\ x^3 \end{pmatrix}$ and in a single equation: $\bar{x}^\mu = \sum_{\nu=0}^{3} (\Lambda^\mu_\nu) x^\nu,$ with Λ the Lorentz transformation matrix
4D scalar product	Define a 4-vector as any set of four components that transform same as (x^0,x^1,x^2,x^3) under Lorentz transformation: $\bar{a}^\mu = \sum_{\nu=0}^{3} \Lambda^\mu_\nu a^\nu.$ there is a 4-vector analog to the dot product, but the zeroth component has a minus: $-a^0b^0 + a^1b^1 + a^2b^2 + a^3b^3.$ > this is the *four-dimensional scalar product* This has the same value in all inertial systems: $-\bar{a}^0\bar{b}^0 + \bar{a}^1\bar{b}^1 + \bar{a}^2\bar{b}^2 + \bar{a}^3\bar{b}^3 = -a^0b^0 + a^1b^1 + a^2b^2 + a^3b^3;$ > its invariant under Lorentz transformations
covariant vector a_μ	= same as the contravariant a^μ only with a sign for the zeroth component: $a_\mu = (a_0, a_1, a_2, a_3) \equiv (-a^0, a^1, a^2, a^3).$ formally, the Minkowski metric: $a_\mu = \sum_{\nu=0}^{3} g_{\mu\nu} a^\nu, \quad \text{where} \quad g_{\mu\nu} \equiv \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ the scalar product can now be written as: $\sum_{\mu=0}^{3} a^\mu b_\mu,$

4-vector product with itself	The scalar product of a 4-vector with itself can be pos, neg or zero: $a^\mu a_\mu = -(a^0)^2 + (a^1)^2 + (a^2)^2 + (a^3)^2.$ define: $a^\mu a_\mu > 0, \quad a^\mu$ is called **spacelike**. $a^\mu a_\mu < 0, \quad a^\mu$ is called **timelike**. $a^\mu a_\mu = 0, \quad a^\mu$ is called **lightlike**.
displacement 4-vector	= difference between even A at (x^0_A,x^1_A,x^2_A,x^3_A) and B at (x^0_B,x^1_B,x^2_B,x^3_B) $\Delta x^\mu \equiv x^\mu_A - x^\mu_B,$
invariant interval	= scalar product of Δx^μ $I \equiv (\Delta x)^\mu (\Delta x)_\mu = -(\Delta x^0)^2 + (\Delta x^1)^2 + (\Delta x^2)^2 + (\Delta x^3)^2 = -c^2t^2 + d^2$ with t the time difference between two events d the spatial separation >> when you transform to a moving system, the time between A and B is altered, and so is their spatial separation, however I remains the same
space-time diagram	= diagram with position at the x-axis and time on the y-axis > the velocity is the reciprocal of the slope of the curve now: a particle at rest is represented as a vertical line a particle going at the speed of light has a slope of 45° > call the trajectory of a particle the *world line* 
forward/backward light cone	The trajectory of a particle is restricted by two lines at 45° due to speed of light > at time t=0 we can draw two cones: - forward light cone > all the positions you can go - backward light cone > all the positions you might have come from 
slope and space/time/lightlike	the slope of the line connecting two events on a space-time diagram tells you: - displacement is timelike: slope greater than 1 - displacement is spacelike: slope is less than 1 - displacement is lightlike: slope is 1
hyperbola and Lorentz transformations	Under Lorentz transformations the interval $I=x^2-c^2t^2$ is preserved > the locus of all points with a given value I is a hyperbola Now when the displacement is: - timelike: we have a hyperboloid of two sheets - spacelike: we have a hyperboloid of one sheet >> if you perform a Lorentz transform the new coords will lie on the same hyperbola
causality in relativity	If the displacement 4-vector between two events is timelike, their ordering is absolute if the interval is spacelike, their ordering depends on the inertial system from which they are observed > an event on the upper sheet of a timelike hyperboloid definitely occurred after (0,0) lower before 

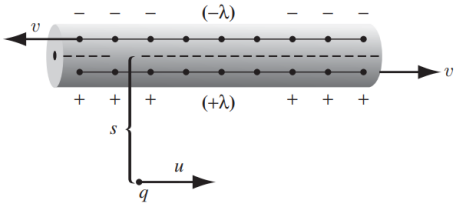
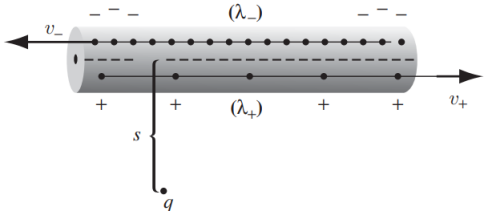
12.2 relativistic mechanics	
12.2.1 proper time and proper velocity	
proper time	as you progress along your world line, your watch runs slow > while the clock on the wall ticks off an interval dt, you watch only advances dτ $$d\tau = \sqrt{1 - u^2/c^2}\, dt.$$
proper velocity	= distance covered per unit proper time: $$\boldsymbol{\eta} \equiv \frac{d\mathbf{l}}{d\tau}$$
ordinary velocity	call u the ordinary velocity, where u and η are related by: $$\boldsymbol{\eta} = \frac{1}{\sqrt{1 - u^2/c^2}}\mathbf{u}.$$
proper velocity 4-vector / 4-velocity	η is the spatial part of a 4-vector: $\eta^\mu \equiv \frac{dx^\mu}{d\tau}$ whose zeroth component is: $$\eta^0 = \frac{dx^0}{d\tau} = c\frac{dt}{d\tau} = \frac{c}{\sqrt{1 - u^2/c^2}}$$ thus, when you go from system S to S' moving at speed v along the xx'-axis: $$\bar{\eta}^0 = \gamma(\eta^0 - \beta\eta^1),$$ $$\bar{\eta}^1 = \gamma(\eta^1 - \beta\eta^0),$$ $$\bar{\eta}^2 = \eta^2,$$ $$\bar{\eta}^3 = \eta^3.$$ or more generally: $\bar{\eta}^\mu = \Lambda^\mu_\nu \eta^\nu;$
12.2.2 relativistic energy and momentum	
relativistic momentum	define momentum using the proper velocity: $$\mathbf{p} \equiv m\boldsymbol{\eta} = \frac{m\mathbf{u}}{\sqrt{1 - u^2/c^2}}$$ which is the *spatial* part of a 4-vector $$p^\mu \equiv m\eta^\mu,$$
relativistic energy	the rel momentum is the spatial part of a 4-vector > the temporal component is the energy: $p^0 = m\eta^0 = \frac{mc}{\sqrt{1 - u^2/c^2}}$ thus: $$E \equiv \frac{mc^2}{\sqrt{1 - u^2/c^2}};$$ thus we have an energy-momentum 4-vector p^μ:
> rest energy	the relativistic energy is nonzero, even if the object is stationary: $E_{\text{rest}} \equiv mc^2.$
> kinetic energy	= the energy which isn't rest energy: $$E_{\text{kin}} \equiv E - mc^2 = mc^2\left(\frac{1}{\sqrt{1 - u^2/c^2}} - 1\right)$$
conservation of momentum and energy	In every isolated system, the total relativistic energy and momentum are conserved > mass is NOT conserved
difference invariant-conserved	Invariant = same value in all inertial systems conserved = same value before and after some process > mass is invariant but not conserved, energy is conserved but not invariant, electric charge is conserved and invariant, velocity is neither conserved nor invariant
relation E-p-m	The scalar product of p^μ with itself gives: $p^\mu p_\mu = -(p^0)^2 + (\mathbf{p}\cdot\mathbf{p}) = -m^2c^2,$ thus: $E^2 - p^2c^2 = m^2c^4.$

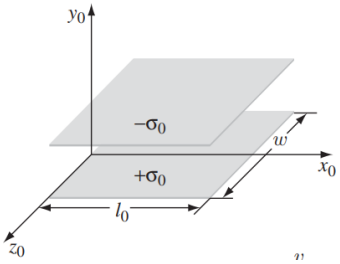
12.2.3 relativistic kinematics	
massless particles	if a particle goes at the speed of light, the denominator in the formulas of E and p is zero > to balance this the mass of the particle is zero > E and p are indeterminate > then the relation holds: $E = pc.$
elastic collisions	Consider a collision where the kinetic energy is conserved, ie elastic > the rest energy is also conserved > the mass is therefore also conserved > the same particles come out as went in
12.2.4 relativistic dynamics	
relativistic force	It still holds: $\mathbf{F} = \frac{d\mathbf{p}}{dt}$ using the relativistic momentum
work-energy theorem	like classical mechanics: $W \equiv \int \mathbf{F} \cdot d\mathbf{l}.$ So: $W = \int \frac{d\mathbf{p}}{dt} \cdot d\mathbf{l} = \int \frac{d\mathbf{p}}{dt} \cdot \frac{d\mathbf{l}}{dt}\, dt = \int \frac{d\mathbf{p}}{dt} \cdot \mathbf{u}\, dt,$ while $\frac{d\mathbf{p}}{dt} \cdot \mathbf{u} = \frac{d}{dt}\left(\frac{m\mathbf{u}}{\sqrt{1-u^2/c^2}}\right) \cdot \mathbf{u}$ $= \frac{m\mathbf{u}}{(1-u^2/c^2)^{3/2}} \cdot \frac{d\mathbf{u}}{dt} = \frac{d}{dt}\left(\frac{mc^2}{\sqrt{1-u^2/c^2}}\right) = \frac{dE}{dt},$ so: $W = \int \frac{dE}{dt}\, dt = E_{\text{final}} - E_{\text{initial}}.$
relativistic 3rd law of Newton	The 3rd law says that two object exert equal force on each other > this does NOT hold nl: say two objects hit each other > in this reference frame, the law still holds however: for a moving observer, these forces will occur at different times
Force in different frame	if we want to calculate the force in a different reference frame, we need to account for both the numerator and denominator: $\bar{F}_y = \frac{d\bar{p}_y}{d\bar{t}} = \frac{dp_y}{\gamma\, dt - \frac{\gamma\beta}{c}\, dx} = \frac{dp_y/dt}{\gamma\left(1 - \frac{\beta}{c}\frac{dx}{dt}\right)} = \frac{F_y}{\gamma(1-\beta u_x/c)}, \quad (12.65)$ and similarly for the z component: $\bar{F}_z = \frac{F_z}{\gamma(1-\beta u_x/c)}.$ The x component is even worse: $\bar{F}_x = \frac{d\bar{p}_x}{d\bar{t}} = \frac{\gamma\, dp_x - \gamma\beta\, dp^0}{\gamma\, dt - \frac{\gamma\beta}{c}\, dx} = \frac{\frac{dp_x}{dt} - \beta\frac{dp^0}{dt}}{1 - \frac{\beta}{c}\frac{dx}{dt}} = \frac{F_x - \frac{\beta}{c}\left(\frac{dE}{dt}\right)}{1-\beta u_x/c}.$ We calculated dE/dt in Eq. 12.63; putting that in, $\bar{F}_x = \frac{F_x - \beta(\mathbf{u}\cdot\mathbf{F})/c}{1-\beta u_x/c}. \quad (12.66)$

Minkowski force	We could also use the same principle as proper time to form less ugly equations: $$K^\mu \equiv \frac{dp^\mu}{d\tau}.$$ with spatial components: $$\mathbf{K} = \left(\frac{dt}{d\tau}\right)\frac{d\mathbf{p}}{dt} = \frac{1}{\sqrt{1-u^2/c^2}}\mathbf{F},$$ and the zeroth component: $$K^0 = \frac{dp^0}{d\tau} = \frac{1}{c}\frac{dE}{d\tau},$$
centre of energy	in classical mechanics the total momentum is: $$\mathbf{P} = M\frac{d\mathbf{R}_m}{dt}$$ with M the total mass and R_m the centre of mass Now: in relativistic mechanics, we have a centre of energy: $\mathbf{R}_e = \frac{1}{E}\sum E_i \mathbf{r}_i$, > replace M by E/c^2 and we have: $$\mathbf{P} = \frac{E}{c^2}\frac{d\mathbf{R}_e}{dt}.$$ >> now P includes all forms of momentum and E all forms of energy

12.3 relativistic electrodynamics

12.3.1 magnetism as a relativistic phenomenon

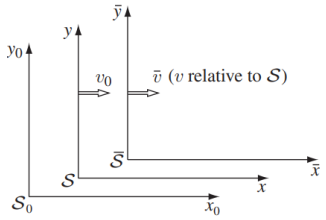
example: wire through current relativistically	suppose we have a string of pos charges moving to the right at speed v, with line charge λ > superimpose a neg string going to the left at same speed v, with line charge -λ > we have a net current going to the right: $$I = 2\lambda v$$  a distance s away is a charge q > traveling to the right at speed u<v > there is no electrical force on q nl: the currents cancel out Now take reference frame Ā moving to the right at speed u, thus q is at rest > the velocities of the neg and pos lines are now: $$v_\pm = \frac{v \mp u}{1 \mp vu/c^2}.$$ now v_- is greater than v_+ > the spacing between the neg charges is closer than the pos charges > the wire carries a net negative charge: with: $$\lambda_\pm = \pm(\gamma_\pm)\lambda_0,$$ $$\gamma_\pm = \frac{1}{\sqrt{1-v_\pm^2/c^2}},$$  where λ_0 is the charge density of the pos line in its own frame Now: both speeds are v in A, thus we can write: $$\lambda = \gamma\lambda_0,$$ We can simplify: $$\gamma_\pm = \frac{1}{\sqrt{1-\frac{1}{c^2}(v\mp u)^2(1\mp vu/c^2)^{-2}}} = \frac{c^2 \mp uv}{\sqrt{(c^2\mp uv)^2 - c^2(v\mp u)^2}}$$ $$= \frac{c^2\mp uv}{\sqrt{(c^2-v^2)(c^2-u^2)}} = \gamma\frac{1\mp uv/c^2}{\sqrt{1-u^2/c^2}}. \quad (1$$ giving us: $$\lambda_{\text{tot}} = \lambda_+ + \lambda_- = \lambda_0(\gamma_+ - \gamma_-) = \frac{-2\lambda uv}{c^2\sqrt{1-u^2/c^2}}.$$ thus: due to length contraction, a neutral wire in one frame is charged in another

force on the charge q	Since in Ā there is a net charge, the λ_{tot} will set up an electrical field $$E = \frac{\lambda_{\text{tot}}}{2\pi\epsilon_0 s}$$ thus the force on q is: $$\bar{F} = qE = -\frac{\lambda v}{\pi\epsilon_0 c^2 s}\frac{qu}{\sqrt{1-u^2/c^2}}$$ however, we should also find a force on q in A > we can find this using transformation rules on forces: $$F = \sqrt{1-u^2/c^2}\,\bar{F} = -\frac{\lambda v}{\pi\epsilon_0 c^2}\frac{qu}{s}.$$ in Ā this is an electrical force, however in A this is definitely NOT an electrical force > this is the magnetic force > namely, if we transform the expression of F a bit, we find: $$F = -qu\left(\frac{\mu_0 I}{2\pi s}\right)$$ which is the expression we would find for the magnetic force in A
12.3.2 how fields transform	
invariance of charge	the charge of a particle is invariant under Lorentz transformations
capacitor relativistically	Consider two charged parallel plates, $-\sigma_0$ and $+\sigma_0$, in a reference frame S_0 at rest > there is a field E_0 produced, perpendicular to the plates: $$\mathbf{E}_0 = \frac{\sigma_0}{\epsilon_0}\hat{\mathbf{y}}.$$  now, in a reference frame S, moving to the right at speed v_0 > the plates are moving to the left at speed v_0 > they still produce a field: $\mathbf{E} = \frac{\sigma}{\epsilon_0}\hat{\mathbf{y}};$ however, the length of the plate is contracted by a factor: $$\gamma_0 = \frac{1}{\sqrt{1-v_0^2/c^2}},$$ and thus the charge per unit area is increased by a factor of γ_0: $$\sigma = \gamma_0\sigma_0.$$ and thus, the field perpendicular to the motion of S: $$\mathbf{E}^{\perp} = \gamma_0\mathbf{E}_0{}^{\perp}$$ but in other dimensions, the plates aren't contracted > so for the field parallel to the plates, we still have: $E^{\parallel} = E_0{}^{\parallel}.$

relativistic transformation of E and B

Consider a moving capacitor in reference frame A:

$$E_y = \frac{\sigma}{\epsilon_0}$$



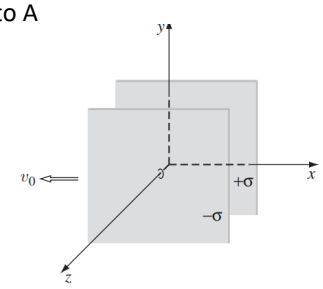
and a magnetic field due to surface currents:

$$\mathbf{K}_{\pm} = \mp\sigma v_0 \hat{\mathbf{x}}.$$

this field points in the negative z direction
> via Ampère we find its magnitude:

$$B_z = -\mu_0 \sigma v_0.$$

Now: consider Ā traveling to the right at speed v in reference to A
> the transformed fields are:



$$\bar{E}_y = \frac{\bar{\sigma}}{\epsilon_0}, \quad \bar{B}_z = -\mu_0 \bar{\sigma} \bar{v},$$

with ¯v¯ the velocity of Ā relative to A_0:

$$\bar{v} = \frac{v + v_0}{1 + v v_0/c^2}, \quad \bar{\gamma} = \frac{1}{\sqrt{1 - \bar{v}^2/c^2}},$$

and:

$$\bar{\sigma} = \bar{\gamma}\sigma_0.$$

Now we want to express ¯**E**¯ and ¯**B**¯ in terms of **E** and **B**
> we have: $\bar{E}_y = \left(\frac{\bar{\gamma}}{\gamma_0}\right)\frac{\sigma}{\epsilon_0}, \quad \bar{B}_z = -\left(\frac{\bar{\gamma}}{\gamma_0}\right)\mu_0\sigma\bar{v}.$

where $\frac{\bar{\gamma}}{\gamma_0} = \frac{\sqrt{1 - v_0^2/c^2}}{\sqrt{1 - \bar{v}^2/c^2}} = \frac{1 + v v_0/c^2}{\sqrt{1 - v^2/c^2}} = \gamma\left(1 + \frac{v v_0}{c^2}\right)$

thus, we can write:

$$\bar{E}_y = \gamma\left(1 + \frac{v v_0}{c^2}\right)\frac{\sigma}{\epsilon_0} = \gamma\left(E_y - \frac{v}{c^2\epsilon_0\mu_0}B_z\right)$$

and:

$$\bar{B}_z = -\gamma\left(1 + \frac{v v_0}{c^2}\right)\mu_0\sigma\left(\frac{v + v_0}{1 + v v_0/c^2}\right) = \gamma(B_z - \mu_0\epsilon_0 v E_y)$$

and since $1/c^2 = 1/\mu_0\varepsilon_0$:

$$\bar{E}_y = \gamma(E_y - v B_z),$$

$$\bar{B}_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$$

Now rotate the capacitor parallel to the xy plane instead of the xz plane
> then we find: $\bar{E}_z = \gamma(E_z + v B_y),$

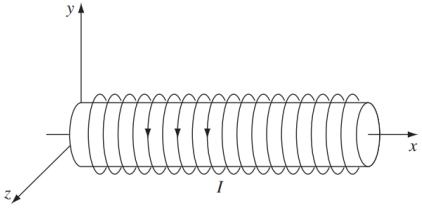
$$\bar{B}_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right)$$

And for the x-component, we have already seen that:

$$\bar{E}_x = E_x$$

however, for the magnetic field in the x-component, we need to do a different experiment

imagine a long solenoid aligned parallel to the x-axis and at rest in A
> magnetic field in the coil is:



$$B_x = \mu_0 n I,$$

with n the number of turns per unit length
> in Ā the length contracts, and n changes:

$$\bar{n} = \gamma n.$$

however time also dilates
> the current in Ā is changed:

$$\bar{I} = \frac{1}{\gamma}I.$$

these cancel out, and we get:

$$\bar{B}_x = B_x$$

transformation rules	$\bar{E}_x = E_x, \quad \bar{E}_y = \gamma(E_y - vB_z), \quad \bar{E}_z = \gamma(E_z + vB_y),$ $\bar{B}_x = B_x, \quad \bar{B}_y = \gamma\left(B_y + \frac{v}{c^2}E_z\right), \quad \bar{B}_z = \gamma\left(B_z - \frac{v}{c^2}E_y\right)$ if **B**=0, then: $\bar{\mathbf{B}} = -\frac{1}{c^2}(\mathbf{v} \times \bar{\mathbf{E}})$ if **E**=0, then: $\bar{\mathbf{E}} = \mathbf{v} \times \bar{\mathbf{B}}$
12.3.3 the field tensor	
tensors	We have Λ the Lorentz transformation matrix: $\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ then a second-rank tensor is an object with two indices > transforms with two factors of Λ: $\bar{t}^{\mu\nu} = \Lambda^{\mu}_{\lambda}\Lambda^{\nu}_{\sigma}t^{\lambda\sigma}$ We can express a rank-2 tensor as a matrix: $t^{\mu\nu} = \begin{Bmatrix} t^{00} & t^{01} & t^{02} & t^{03} \\ t^{10} & t^{11} & t^{12} & t^{13} \\ t^{20} & t^{21} & t^{22} & t^{23} \\ t^{30} & t^{31} & t^{32} & t^{33} \end{Bmatrix}$
symmetric tensor antisymmetric tensor	= tensor for which: $t^{\mu\nu} = t^{\nu\mu}$ = tensor for which: $t^{\mu\nu} = -t^{\nu\mu}$
Field tensor $F^{\mu\nu}$	A general antisymmetric rank-2 tensor has 6 components: $t^{\mu\nu} = \begin{Bmatrix} 0 & t^{01} & t^{02} & t^{03} \\ -t^{01} & 0 & t^{12} & t^{13} \\ -t^{02} & -t^{12} & 0 & t^{23} \\ -t^{03} & -t^{13} & -t^{23} & 0 \end{Bmatrix}$ we can work out the components via: $\bar{t}^{01} = \Lambda^{0}_{\lambda}\Lambda^{1}_{\sigma}t^{\lambda\sigma}$ where we find: $\left.\begin{array}{lll} \bar{t}^{01} = t^{01}, & \bar{t}^{02} = \gamma(t^{02} - \beta t^{12}), & \bar{t}^{03} = \gamma(t^{03} + \beta t^{31}), \\ \bar{t}^{23} = t^{23}, & \bar{t}^{31} = \gamma(t^{31} + \beta t^{03}), & \bar{t}^{12} = \gamma(t^{12} - \beta t^{02}). \end{array}\right\}$ > these are very similar to the transformation rules of electromagnetism > we can construct the field tensor: $F^{\mu\nu} = \begin{Bmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & B_z & -B_y \\ -E_y/c & -B_z & 0 & B_x \\ -E_z/c & B_y & -B_x & 0 \end{Bmatrix}$
Dual tensor $G^{\mu\nu}$	We can substitute E/c→B and B→-E/c without changing the outcome of the tensor: $G^{\mu\nu} = \begin{Bmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & -E_z/c & E_y/c \\ -B_y & E_z/c & 0 & -E_x/c \\ -B_z & -E_y/c & E_x/c & 0 \end{Bmatrix}$

12.3.4 electrodynamics in tensor notation

current density J^μ

consider an infinitesimal volume V a cloud of charge with charge Q going at speed **u**
> the charge density is: $\rho = \frac{Q}{V}$

and the current density is: $\mathbf{J} = \rho\mathbf{u}.$

define the proper charge density ρ_0 as the density in the rest system of the charge:

$$\rho_0 = \frac{Q}{V_0}$$

with V_0 the rest volume of the cloud
> we measure a volume V according to:

$$V = \sqrt{1 - u^2/c^2}\, V_0,$$

therefore:

$$\rho = \rho_0 \frac{1}{\sqrt{1 - u^2/c^2}}, \quad \mathbf{J} = \rho_0 \frac{\mathbf{u}}{\sqrt{1 - u^2/c^2}}.$$

these make a 4-vector:

$$J^\mu = \rho_0 \eta^\mu,$$

with components:

$$J^\mu = (c\rho, J_x, J_y, J_z)$$

continuity equation

We have: $\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t},$

where: $\nabla \cdot \mathbf{J} = \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}$ and $\frac{\partial \rho}{\partial t} = \frac{1}{c}\frac{\partial J^0}{\partial t} = \frac{\partial J^0}{\partial x^0}.$

thus: $\frac{\partial J^\mu}{\partial x^\mu} = 0,$

Maxwell's equations

We have: $\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \mu_0 J^\mu, \quad \frac{\partial G^{\mu\nu}}{\partial x^\nu} = 0,$

namely:, we find gauss's law via:

$$\frac{\partial F^{0\nu}}{\partial x^\nu} = \frac{\partial F^{00}}{\partial x^0} + \frac{\partial F^{01}}{\partial x^1} + \frac{\partial F^{02}}{\partial x^2} + \frac{\partial F^{03}}{\partial x^3}$$

$$= \frac{1}{c}\left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}\right) = \frac{1}{c}(\nabla \cdot \mathbf{E})$$

$$= \mu_0 J^0 = \mu_0 c\rho,$$

<=>

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0}\rho.$$

we can find Ampère's law, namely if μ=1:

$$\frac{\partial F^{1\nu}}{\partial x^\nu} = \frac{\partial F^{10}}{\partial x^0} + \frac{\partial F^{11}}{\partial x^1} + \frac{\partial F^{12}}{\partial x^2} + \frac{\partial F^{13}}{\partial x^3}$$

$$= -\frac{1}{c^2}\frac{\partial E_x}{\partial t} + \frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} = \left(-\frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} + \nabla \times \mathbf{B}\right)_x$$

$$= \mu_0 J^1 = \mu_0 J_x.$$

combine this with the results of μ=2 en μ=3:

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

we can find the third maxwell equation via:

$$\frac{\partial G^{0\nu}}{\partial x^\nu} = \frac{\partial G^{00}}{\partial x^0} + \frac{\partial G^{01}}{\partial x^1} + \frac{\partial G^{02}}{\partial x^2} + \frac{\partial G^{03}}{\partial x^3}$$

$$= \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = \nabla \cdot \mathbf{B} = 0$$

and Faraday's law via: $\frac{\partial G^{1\nu}}{\partial x^\nu} = \frac{\partial G^{10}}{\partial x^0} + \frac{\partial G^{11}}{\partial x^1} + \frac{\partial G^{12}}{\partial x^2} + \frac{\partial G^{13}}{\partial x^3}$

$$= -\frac{1}{c}\frac{\partial B_x}{\partial t} - \frac{1}{c}\frac{\partial E_z}{\partial y} + \frac{1}{c}\frac{\partial E_y}{\partial z} = -\frac{1}{c}\left(\frac{\partial \mathbf{B}}{\partial t} + \nabla \times \mathbf{E}\right)_x = 0.$$

Minkowski's force	we find Minkowski's force via: $$K^\mu = q\eta_\nu F^{\mu\nu}.$$ nl, for μ=1: $K^1 = q\eta_\nu F^{1\nu} = q(-\eta^0 F^{10} + \eta^1 F^{11} + \eta^2 F^{12} + \eta^3 F^{13})$ $$= q\left[\frac{-c}{\sqrt{1-u^2/c^2}}\left(\frac{-E_x}{c}\right) + \frac{u_y}{\sqrt{1-u^2/c^2}}(B_z) + \frac{u_z}{\sqrt{1-u^2/c^2}}(-B_y)\right]$$ $$= \frac{q}{\sqrt{1-u^2/c^2}}[\mathbf{E} + (\mathbf{u}\times\mathbf{B})]_x,$$ which combined with μ=2 and μ=3 gives us: $$\mathbf{K} = \frac{q}{\sqrt{1-u^2/c^2}}[\mathbf{E} + (\mathbf{u}\times\mathbf{B})],$$
12.3.5 relativistic potentials	
potential A	define the vector: $A^\mu = (V/c, A_x, A_y, A_z)$ then: $$F^{\mu\nu} = \frac{\partial A^\nu}{\partial x_\mu} - \frac{\partial A^\mu}{\partial x_\nu}$$
gauge invariance of A	we can add the gradient of any scalar function λ without changing $F^{\mu\nu}$: $$A^\mu \longrightarrow A^{\mu\prime} = A^\mu + \frac{\partial\lambda}{\partial x_\mu}$$ for example the Lorentz gauge: $$\nabla\cdot\mathbf{A} = -\frac{1}{c^2}\frac{\partial V}{\partial t}$$ or in relativistic notation: $$\frac{\partial A^\mu}{\partial x^\mu} = 0.$$
Alamnertian equation of A	We know: $\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \mu_0 J^\mu,$ thus: $\frac{\partial}{\partial x_\mu}\left(\frac{\partial A^\nu}{\partial x^\nu}\right) - \frac{\partial}{\partial x_\nu}\left(\frac{\partial A^\mu}{\partial x^\nu}\right) = \mu_0 J^\mu.$ we can use the Lorentz gauge: $$\frac{\partial A^\mu}{\partial x^\mu} = 0.$$ with which we can write: $$\Box^2 A^\mu = -\mu_0 J^\mu$$ where: $\Box^2 \equiv \frac{\partial}{\partial x_\nu}\frac{\partial}{\partial x^\nu} = \nabla^2 - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}$